WEBSITE DEVELOPMENT / DOCUMENT 03 — ANSWERS

SOFTWARE REQUIREMENTS SPECIFICATION

Completed SRS questionnaire for the RIPS website

PROJECT / WEBSITE	Radiant Institute of Professional Studies (RIPS), Sheikhupura — rips.edu.pk
PREPARED BY	Sarosh (website developer)
DATE	17 October 2026
VERSION	1.0 draft answers — for review
COVERAGE	192 questions · 20 sections · 6 feature specifications · 14 quality requirements · 18 open decisions
FOLLOWS	PVS answers (draft v0.1) and PRD answers (draft v1.0), both dated 17 Oct 2026 and not yet approved

How to read this document. Answers are consistent with the earlier PVS and PRD answers. **Not decided** marks a genuine open decision; each one is listed in the Open decisions register (OD-xx) with an owner. *(proposed)* and *(assumption)* mark numbers or rules that the approver must confirm. **Not applicable** marks features release 1 does not have (for example payments). As the questionnaire asks, technical architecture, database tables and API endpoints are left to later specifications; tool choices are recorded as open decisions.

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A Document Control & Requirement Baseline

Establish the approved sources, decisions, identifiers, and sign-off process.

A-01 What is the exact product name, version, and website URL (if already known)?

> Radiant Institute of Professional Studies (RIPS) website, release 1 (v1.0), at **https://rips.edu.pk**. The official spelling of the institute's name must be confirmed (OD-09).

A-02 Which approved Project Vision & Scope and PRD versions should this SRS follow?

> Project Vision & Scope answers, draft v0.1, 17 Oct 2026; PRD questionnaire answers, draft v1.0, 17 Oct 2026. **Neither is approved yet.** Their approved version numbers and dates must be recorded here before this SRS is signed.

A-03 Who owns this SRS, who reviews it, and who has authority to approve it?

Role	Person
Owner (writes and maintains)	Sarosh, website developer
Reviewers	Director (scope), Principal (academic content), admissions officer (forms, fees, enquiry handling)
Approver	**Not decided** — Chairman or Director (OD-01)

A-04 What is the target first-release date, and are there intermediate milestones?

First-release date: **Not decided** (OD-02); target is before the next admissions intake opens.

Milestone	Target
M0 Holding page live on rips.edu.pk; hacked server cut off	Immediately
M1 SRS approved	Before development starts
M2 Staging site complete (password-protected)	**Not decided**
M3 Closed review by Director, Principal, admissions	About 2 weeks after M2 (proposed)
M4 Public launch v1.0	Before next intake

A-05 Which requirements, features, or decisions from earlier documents are already fixed?

- Domain rips.edu.pk, managed at DOMAIN.pk.
- Old WordPress site and Hostinger hosting are retired; nothing is reused from them.
- Site is built in-house with VS Code.
- Release 1 is an informational site with enquiry; no visitor accounts, payments, student portal, LMS or app.
- Mobile-first; English is the primary language.
- Must-have feature list from PRD Q34.

A-06 How should requirements be prioritized (for example Must/Should/Could/Won't)?

> MoSCoW. **Must** = launch is blocked without it. **Should** = in release 1 unless it endangers the date. **Could** = only if time allows. **Won't** = not in release 1, recorded for later.

A-07 Who may request changes, and who approves changes after SRS sign-off?

> Anyone in the stakeholder list (A-03) may request a change by message or email to the developer. The developer writes down the impact (time, risk, other requirements). The approver (OD-01) accepts or rejects it in writing. See T-10.

A-08 What existing specifications, policies, designs, or regulations must be cross-referenced?

> - PVS and PRD answers (A-02).
> - Affiliating university and professional-council rules on how programs are advertised *(confirm documents)*.
> - Institute fee and refund policy, if one exists.
> - WCAG 2.1 AA (accessibility).
> - Pakistan's Prevention of Electronic Crimes Act, 2016, for published content; data-protection obligations to be checked by legal counsel (OD-15).
> - No existing design files or brand guidelines exist; only the logo.

A-09 How should unresolved issues and assumptions be documented and tracked?

> Every open item gets an ID (OD-xx) in the **Open decisions register** at the end of this document, with an owner, due point and status (Open, Decided, Dropped). Assumptions are marked *(assumption)* or *(proposed)* in the text. The register is reviewed at each weekly meeting.

B System Scope & Boundaries

Specify exactly what the system is required to do and what it will not do.

B-01 What are the exact in-scope capabilities for release 1?

> 1. Public pages: Home, About (incl. affiliation and approvals), Programs list and program pages, Admissions (process, eligibility, fees, dates), Notices and merit lists, News and events, Faculty, Contact, Privacy policy.
> 2. Enquiry form with email notification to admissions and an auto-reply to the visitor.
> 3. WhatsApp click-to-chat, tap-to-call, embedded map.
> 4. Downloadable PDFs (prospectus, fee structure, notices, merit lists).
> 5. Staff sign-in and content updates (notices, merit lists, news, dates, fees).
> 6. Staff view, status-tracking and export of enquiries.
> 7. Search-engine optimisation (titles, sitemap, structured data) and privacy-safe analytics.

B-02 What capabilities are explicitly out of scope for release 1?

> Online payments; visitor or student accounts; online application with document upload; Open Day booking; student portal (results, attendance, fee vouchers); LMS; mobile app; live chat; public comments or reviews; hostel or transport booking; SMS messaging; full Urdu version (key Urdu pages are a Could).

B-03 Which features are read-only and which allow creating, editing, or deleting data?

Feature	Visitors	Staff
All public pages and PDFs	Read-only	Create, edit, archive, delete (by role)
Enquiry form	Create only	Read, change status, export, delete on request
Staff accounts	—	Administrator creates, edits, disables, deletes

B-04 Which operations must be available publicly versus only after authentication?

Public: view every public page, download public PDFs, submit enquiry and contact forms, open WhatsApp, call, map. **Authenticated (staff only):** everything that changes content, reading enquiries, exports, user management, settings.

B-05 What will remain a manual staff or offline process rather than website functionality?

- Calling and WhatsApp-ing enquirers back.
- Applications, document checks and fee payment (on campus or by bank, as now).
- Preparing merit lists (the site only publishes the PDF).
- Translating content into Urdu.
- Approving fees, dates and notices (done by people; the site only records who published).

B-06 Which third-party systems are inside the operational workflow but outside your control?

Domain registrar DOMAIN.pk and the .pk registry; the new hosting provider (OD-03); email provider (OD-04); form-and-email service (OD-06); WhatsApp; Google Maps; Google Analytics and Search Console; YouTube.

B-07 Are there separate websites, portals, subdomains, or mobile web experiences?

Release 1 has one public site (rips.edu.pk, with www redirecting to it) and one staff area. A staging copy lives on a separate, password-protected subdomain (proposed staging.rips.edu.pk). **test.rips.pk** is an older, public copy on a different domain: its ownership must be confirmed and it must be taken down or hidden (OD-17). No separate mobile site; the same site adapts to phones.

B-08 What major constraints or dependencies could prevent any requirement from being delivered?

- Late or unapproved content (fees, eligibility, program list, affiliation papers).
- No decision on host, email or admin tool.
- DNS change at DOMAIN.pk not possible (account access).
- One developer only.
- Admission-season deadline.
- Institute email breaking during the DNS switch if MX records are not recorded first.

C Users, Accounts & Permissions

Identify actors and define security-sensitive access rules.

C-01 List every actor or user role, including guests, customers, staff, administrators, and external systems.

Actor	Type
Visitor (prospective student, parent, current student, public)	Guest, no account
Admissions officer	Staff account
Content editor	Staff account
Site administrator	Staff account
Reviewer (Principal or Director)	Approves outside the system in release 1; optional read-only account
Form-and-email service	External system
Search engines and analytics	External systems

C-02 What actions may each role view, create, update, approve, export, or delete?

Object	Visitor	Admissions officer	Content editor	Administrator
Public pages	View	Edit admission pages, fees, dates	Edit programs, about, faculty, campus	All
Notices, merit lists	View	Create, update, archive, delete	Create, update notices (not merit lists)	All
News, events, gallery	View	—	Create, update, archive, delete	All
Enquiries	Create	View, update status, export, delete	—	View, export, delete
Staff accounts, settings	—	Own profile	Own profile	Create, update, disable, delete

Approval of fees, dates and merit lists happens before publishing and is recorded by the publisher (L-04).

C-03 Can an individual hold multiple roles? If so, how is the effective permission determined?

Yes. Effective permission is the **union** of the roles held. Only the administrator assigns roles. At least two people should hold the administrator role so the site is never locked out (proposed).

C-04 How can users register, be invited, or be created by an administrator?

No self-registration. The administrator creates a staff account or sends an invitation link to the person's institute email address. Visitors never register in release 1.

C-05 Which registration fields are mandatory and which are optional?

Staff account: full name, institute email address and role are mandatory; phone number is optional.

C-06 How is each account verified or activated (email, SMS, staff approval, or other)?

> The invited person opens the emailed link (valid 72 hours), sets a password and sets up two-factor sign-in. The account is active only after both steps. Creation itself is the staff approval.

C-07 Which login methods are supported, and is multifactor authentication required for any role?

> Email and password. Two-factor authentication (authenticator app or emailed code) is **required for administrators** and should be required for all staff if the chosen admin tool supports it (OD-05). No social sign-in.

C-08 What is the exact password reset, account recovery, and change-email/phone process?

> 1. Staff clicks “Forgot password” and enters their email.
> 2. The system always replies “If this email has an account, a reset link has been sent” (does not reveal whether the account exists).
> 3. The link is single-use and expires after 60 minutes.
> 4. New password: at least 12 characters, not the same as the last one.
> 5. All other sessions are signed out after a reset.
>
> Changing email address: only the administrator can do it, and a notice goes to both the old and the new address. Lost two-factor device: the administrator resets two-factor after confirming identity in person or by phone.

C-09 When should sessions expire, and must users be able to sign out from all devices?

> Staff sessions end after 30 minutes without activity or 8 hours in total (proposed). Staff can sign out. The administrator can end all sessions for an account (Should; depends on the admin tool).

C-10 Can accounts be disabled, suspended, reinstated, or deleted, and by whom?

> Only the administrator. Accounts are **disabled** (not deleted) when someone leaves, the same day; disabled accounts can be reinstated. Deletion happens after 90 days of being disabled; the activity log keeps the person’s name.

C-11 What does a user see or receive when access is denied or an account is suspended?

> Wrong credentials: “Email or password is incorrect.” Disabled account: “This account is not active. Please contact the site administrator.” Visitor opening a staff address: the sign-in page; no staff content is ever shown. Access denied for a role: “You do not have permission to do this.”

D Pages, Navigation & Visible States

Describe observable behavior of every page, not just its visual layout.

D-01 What is the complete list of pages or screens for release 1?

Public pages (proposed address)	Staff screens
Home (/)	Sign in, forgot password, set password
About (/about), Affiliation and approvals (/about/affiliation)	Dashboard
Programs list (/programs), one page per program (/programs/[name])	Enquiries list and enquiry detail

Public pages (proposed address)	Staff screens
Admissions (/admissions), Fee structure (/admissions/fees)	Notices and merit lists list, add/edit
Notices and merit lists (/notices), notice detail	News and events list, add/edit
News and events (/news), post page	Pages editor (programs, admissions, about, faculty)
Faculty (/faculty)	Media and files
Campus and facilities (/campus) — Could	Users and roles (administrator only)
Careers (/careers) — Could	Activity log (administrator only)
Contact (/contact), Thank-you (/thank-you)	My profile
Privacy policy (/privacy), Terms (/terms), Not found (404)	

D-02 Which roles can reach each page or navigation item?

Every public page: everyone. Staff screens: any signed-in staff member, filtered by role per C-02. Users and roles and Activity log: administrator only. Enquiries: admissions officer and administrator.

D-03 What is the expected default landing page for each user role?

Role	Lands on
Visitor	Home
Admissions officer	Enquiries list, newest first
Content editor	Content list (drafts first)
Administrator	Dashboard

D-04 For each page, what information, actions, and user-facing outputs must appear?

As listed in PRD Q45, summarised:

Page	Must show	Actions
Home	Name, logo, one-line value, program cards, admission dates strip, latest 5 notices, contact strip	Enquire, WhatsApp, Programs, Admissions
Program page	Name, level, duration, eligibility, fee, affiliation, outline, careers, intake status badge, last updated	Enquire (program pre-selected), download details
Admissions	Steps, documents needed, dates, fee table (PKR), scholarships, last updated	Download prospectus and fee PDF, Enquire
Notices	Title, date, program, session, type, file size	Filter, open PDF
Contact	Address, phone, WhatsApp, email, office hours, map	Call, WhatsApp, directions, contact form
Thank-you	Confirmation text, what happens next, WhatsApp link	Back to programs

D-05 How should navigation, back buttons, breadcrumbs, and redirects behave?

- The browser Back button always returns to the previous page and position; no page breaks Back.
- Breadcrumbs on program, notice and news pages (e.g. Home › Programs › Pharm-D).
- http → https and www → rips.edu.pk with permanent (301) redirects.
- Known old addresses from the WordPress site (found from Google search results) redirect permanently to the nearest new page.
- After the enquiry form succeeds, the visitor goes to the Thank-you page.
- After staff sign-in, the role's landing page (D-03).

D-06 What should happen if a user visits an invalid, removed, or restricted URL?

Invalid: a 404 page "Sorry, we couldn't find that page" with links to Programs, Admissions, Contact. Removed notice: a 410-style page "This notice is no longer available" with a link to current notices. Restricted: redirect to staff sign-in; after sign-in, return to the requested page if the role allows it, otherwise "You do not have permission".

D-07 What empty, loading, success, warning, and failure states must each major page support?

State	Behaviour
Empty	Notices: "Nothing posted yet." News: "No news yet." Enquiries (staff): "No enquiries yet."
Loading	Pages are pre-built and load at once; form submit shows a spinner and "Sending…" and disables the button
Success	Form: Thank-you page. Staff save: "Saved" / "Published" message
Warning	Intake closed badge; "Fees may change — last updated [date]"; unsaved staff changes: "You have unsaved changes. Leave anyway?"
Failure	Form: "We couldn't send your enquiry. Please try again, or WhatsApp us at [number]." Staff save failure: "Not saved — please try again", with the typed content kept

D-08 What should be preserved if a user refreshes, navigates away, or returns to a page?

Filter choices on Programs and Notices are kept in the address, so refresh and sharing keep them. Form inputs are kept after a validation error. Nothing is saved after the visitor leaves the form page. Staff editors warn before leaving unsaved changes.

D-09 Do any pages need print views, downloadable versions, or shareable links?

Print-friendly layout for program pages, admissions and the fee table (no menus or buttons printed). Downloadable PDFs for prospectus, fee structure, notices and merit lists. Every program, notice and news post has a stable, shareable address with a link preview image for WhatsApp and Facebook.

E Feature-by-Feature Functional Requirements

Answer these for every major feature, using the repeatable worksheet at the end.

E-01 What is the unique name and purpose of each core feature?

ID	Name	Purpose
FEAT-001	Program pages	Show what each program offers so visitors can decide and enquire
FEAT-002	Admissions and fees	Show steps, dates, documents and fees for the current intake
FEAT-003	Enquiry form	Capture a visitor's details and deliver them to admissions
FEAT-004	Notices and merit lists	Publish official, dated updates as PDFs
FEAT-005	WhatsApp, call and contact	One-tap contact with admissions and directions
FEAT-006	Staff content and enquiry management	Let staff publish content and track enquiries without the developer

Each feature has a full Worksheet 1 at the end of this document.

E-02 Which actor can start each feature, and what event triggers it?

Feature	Actor	Trigger
FEAT-001, 002	Visitor	Opens the page from menu, search or shared link
FEAT-003	Visitor	Taps Enquire or opens Contact
FEAT-004	Visitor (read); admissions officer or editor (publish)	Opens Notices; staff clicks "Add notice"
FEAT-005	Visitor	Taps WhatsApp, call or map
FEAT-006	Staff	Signs in

E-03 What are the required preconditions before the feature may run?

Feature	Preconditions
FEAT-001	Program approved and affiliation confirmed; content entered
FEAT-002	Current intake dates and fees approved
FEAT-003	Form service configured; admissions mailbox working; privacy policy published
FEAT-004	Approved PDF ready; staff signed in with the right role
FEAT-005	Staffed WhatsApp number, phone and verified address configured
FEAT-006	Staff account active with two-factor set up

E-04 What exact steps must the website perform in the normal success scenario?

FEAT-003 (the only feature that processes input) as the example; the others are in the worksheets:

1. Show the form with the program pre-selected if opened from a program page.
2. Validate each field when the visitor leaves it.
3. On Submit, validate everything again and check the spam guard.
4. Disable the button and show “Sending…”.
5. Send the enquiry to the form service.
6. On success, store the enquiry with status New, email admissions, email the visitor an auto-reply if an email was given.
7. Show the Thank-you page.

E-05 What are the expected results and outputs after successful completion?

Feature	Output
FEAT-001, 002	Page shown with all required fields; PDFs download
FEAT-003	Thank-you page; enquiry record (New); email to admissions within 1 minute; auto-reply
FEAT-004	Notice visible on the public list within 5 minutes, newest first
FEAT-005	WhatsApp opens with pre-filled message; dialer opens; map opens
FEAT-006	Change published or saved; activity logged

E-06 What alternative paths must be supported when users take different valid actions?

- Enquiry opened from the header (no program pre-selected) — visitor picks one, or “Not sure yet”.
- Visitor chooses WhatsApp instead of the form.
- Staff saves a notice as Draft instead of publishing.
- Staff schedules a notice to publish at a future date and time (Could).
- Visitor filters notices by program or type.

E-07 What business decisions or calculations must the system perform?

- Intake status badge from dates: Upcoming (before opening date), Open, Closing soon (7 days or less before the closing date), Closed (after closing date).
- Notices move to Archive automatically after their archive date (default: 30 days after the intake closes).
- Enquiries flagged as possible duplicates when the same phone number and program arrive within 7 days.
- No fee calculations in release 1; fees are displayed as entered.

E-08 Which inputs are required and which defaults should be applied?

Field rules are in F-02. Defaults: program = the page the visitor came from, otherwise none; enquiry status = New; notice date = today; notice program = All programs; notice archive date = 30 days after intake closes; content status = Draft.

E-09 What data or status changes must occur when this feature succeeds?

Feature	Change
FEAT-003	New enquiry record with status New and received time (Pakistan time)
FEAT-004	Notice status Draft → Published; publish time and publisher recorded
FEAT-006	Content version saved; activity log entry; enquiry status changes recorded with who and when

E-10 What must happen when a feature is cancelled, timed out, retried, or interrupted?

- Visitor closes the form: nothing is stored.
- Form submission takes over 15 seconds: show the failure message with WhatsApp and phone; typed values stay; visitor can retry.
- Retry after a network error must not create a second enquiry if the first one was received (Q-03).
- Staff session times out while editing: the edit is kept in the browser and restored after sign-in where the tool allows; otherwise the warning in D-07 applies.

E-11 Which feature actions must require explicit confirmation, approval, or verification?

- Publishing or replacing a merit list: confirmation dialog that names the program and session.
- Changing fees or dates: confirmation, plus approval recorded (approver's name entered).
- Deleting any content or enquiry: "Delete [title]? This cannot be undone."
- Exporting enquiries: confirmation that the file contains personal data.
- Visitor consent checkbox before an enquiry is accepted.

E-12 What are the exact measurable acceptance criteria for each feature?

Given / When / Then tests for each feature are in Worksheet 1. Common criteria: works at 360 px width on Android Chrome and iOS Safari; passes the accessibility checks in T-04; Lighthouse mobile performance 85 or higher on the feature's page.

F Forms, Input Validation & File Handling

Make entry rules precise, testable, and understandable to end users.

F-01 Which forms exist, and which roles can view or submit each form?

Form	Who can submit
Enquiry form	Any visitor
Contact form	Any visitor
Problem report (footer link)	Any visitor (Could; may be a plain email link)
Staff sign-in, forgot password, set password	Staff
Notice / merit list form	Admissions officer, content editor (notices only), administrator
News and event form, page editor	Content editor, administrator
Enquiry status update	Admissions officer, administrator
User form	Administrator

F-02 For each field, what is its name, type, mandatory status, default, and allowed format?

Enquiry form

Field	Type	Required	Default	Format
Full name	Text	Yes	—	Letters, spaces, . ’ -
Mobile / WhatsApp number	Phone	Yes	—	03XXXXXXXXX or +923XXXXXXXXX (spaces and dashes ignored)
Email	Email	No	—	Standard email address
Program of interest	Dropdown	Yes	Page the visitor came from	Published programs + “Not sure yet”
City	Dropdown + text	Yes	Sheikhupura	List of nearby cities + “Other” (then text)
Last qualification	Dropdown	No	—	Matric/O-level, FSc Pre-Medical, FSc Pre-Engineering, ICS, FA, ICom, A-level, Other
Marks (%)	Number	No	—	0–100, up to 1 decimal
Message	Long text	No	—	Plain text
Consent	Checkbox	Yes	Unticked	Must be ticked
Spam guard	Hidden	—	Empty	Must stay empty
Source page and campaign	Hidden	—	Set automatically	Page address and campaign tag; no personal data

Contact form: name (required), phone or email (one required), subject (optional), message (required), consent (required).

Notice form: title (required), type (Admission, Merit list, Exam, General; required), program (default All programs), session (e.g. 2026–27; required for merit lists), publish date (default today), archive date (default 30 days after intake close), PDF file (required), status (Draft or Published).

F-03 What are the minimum/maximum lengths, numerical ranges, and permitted characters?

Field	Rule
Full name	3–60 characters; letters (incl. Urdu), spaces, . ’ -
Mobile number	11 digits starting 03, or +92 followed by 10 digits starting 3
Email	Up to 100 characters
City (Other)	2–40 characters, letters and spaces
Marks	0–100
Message	Up to 500 characters; links are allowed but more than 2 links marks it as possible spam
Notice title	5–120 characters
Staff password	At least 12 characters

F-04 Which fields depend on selections or values entered in other fields?

- City = “Other” shows the city text box, which then becomes required.
- Marks is shown only after a qualification is chosen.
- Notice type = Merit list makes Session required and limits who can publish (admissions officer, administrator).
- Contact form: at least one of phone or email.

F-05 What duplicate, inconsistent, or contradictory entries must be rejected?

- An archive date earlier than the publish date (staff): rejected.
- Marks outside 0–100: rejected.
- Spam-guard field filled: silently discarded.
- The same phone number and program within 7 days: accepted but flagged “Possible duplicate” for admissions (not rejected, so real follow-ups are not lost).
- A second merit list with the same program and session: staff must choose to replace the existing one or cancel.

F-06 When should validation run: while typing, on field exit, or on submission?

On field exit, and again on submission (in the browser and on receipt). After a field has shown an error, it re-checks while typing so the error clears as soon as the input is fixed.

F-07 What specific user-facing validation message should appear for each invalid input?

Problem	Message
Name empty or too short	Please enter your full name.
Name has invalid characters	Please use letters only in your name.
Mobile empty or wrong format	Please enter a mobile number like 0300 1234567.
Email wrong format	Please check your email address, e.g. name@gmail.com.

Problem	Message
Program not chosen	Please choose a program, or “Not sure yet”.
City empty	Please choose your city.
Marks out of range	Marks must be between 0 and 100.
Message too long	Please keep your message under 500 characters.
Consent not ticked	Please tick the box so we can contact you.
Form-level	Please fix the highlighted fields.

Urdu versions are added when Urdu pages exist.

F-08 Can users save drafts, and how long must unfinished entries remain available?

Visitors: no drafts. Staff: content can be saved as Draft and stays until published or deleted; drafts untouched for 90 days are listed for clean-up.

F-09 What files can users upload, including permitted types, maximum size, count, and limits?

Visitors: none in release 1.

Staff upload	Types	Max size	Count
Notice, merit list, prospectus, fee structure	PDF	5 MB	1 per notice
Images (news, gallery, faculty, programs)	JPG, PNG, WebP	5 MB before automatic compression to about 1 MB	Up to 20 per upload
Affiliation and approval letters	PDF or JPG	5 MB	1 per document

F-10 What checks, previews, replacements, downloads, and removals are required for uploads?

- Check the real file type, not just the name; reject anything else.
- Scan for malware where the host or tool supports it.
- Show a preview (first page of a PDF, or the image) before publishing.
- Replacing a file keeps the same public address and records the old version.
- Removing a published file asks for confirmation and shows the notice as “no longer available”.
- Filenames are cleaned (letters, numbers, dashes).

F-11 How should partial uploads, failed submissions, and duplicate submit clicks be handled?

A partial or failed upload is discarded and the staff member sees “Upload failed — please try again.” A failed visitor submission keeps the typed values and shows the failure message (D-07). The Submit button is disabled after the first click until a result arrives; each submission carries a one-time token so a repeat is ignored.

G Business Rules, Statuses & Lifecycle

Define precise rules and transitions for important business objects.

G-01 What are the main business objects (for example orders, bookings, requests, reports)?

> Enquiry; Contact message; Program; Intake (dates); Fee schedule; Notice (incl. merit list); News or event post; Page; Staff user.

G-02 What statuses can each object have, and what does each status mean?

Object	Statuses
Enquiry	**New** (received, not yet contacted) · **Contacted** (first call or message made) · **Follow-up** (interested, next step pending) · **Visited/Applied** · **Enrolled** · **Not interested** · **Spam/Duplicate**
Notice, news, page	**Draft** · **In review** · **Published** · **Archived** (visible in archive only) · **Deleted**
Intake	**Upcoming** · **Open** · **Closing soon** · **Closed** (calculated from dates, E-07)
Program	**Active** (shown) · **Not offered this intake** (shown with badge) · **Withdrawn** (hidden)
Staff user	**Invited** · **Active** · **Disabled**

G-03 What events or permissions allow each status transition?

Transition	Who / what
Enquiry: New → Contacted → Follow-up → Visited/Applied → Enrolled	Admissions officer or administrator
Enquiry: any → Not interested or Spam/Duplicate	Admissions officer or administrator
Content: Draft → In review → Published	Author submits; publisher with the right role publishes
Content: Published → Archived	Automatically on archive date, or by publisher
Intake statuses	Automatic from dates
Staff: Invited → Active	Person sets password and two-factor
Staff: Active ↔ Disabled	Administrator

G-04 Which transitions are prohibited, and what message appears if one is attempted?

- Enrolled → New: not allowed. “An enrolled enquiry cannot be reset. Add a note instead.”
- Deleted content cannot be published again: “This item was deleted. Create a new one.”
- A content editor publishing a merit list: “Only admissions can publish merit lists.”
- An administrator disabling the last active administrator: “At least one administrator must stay active.”

G-05 Who may edit or withdraw an item after submission or approval?

> Visitors cannot edit an enquiry after sending; they contact admissions. Admissions officer and administrator can add notes and change status; the original enquiry text is never edited. Published content may be edited by its role owner; fee, date and merit-list edits need re-approval (L-04).

G-06 Which deadlines, expiry rules, waiting periods, or time zones apply?

> - All dates and times in Pakistan Standard Time (UTC+5), shown as 17 Oct 2026, 2:30 pm.
> - Admissions responds to enquiries within 1 working day (Monday–Saturday, hours to confirm).
> - Duplicate window: 7 days.
> - Password reset link: 60 minutes. Invitation link: 72 hours.
> - Notices archive 30 days after intake close by default.
> - Enquiries deleted 12 months after the intake they relate to (proposed, OD-10).

G-07 What should happen when two users edit or approve the same item simultaneously?

> The second person to save is warned: “Someone else changed this item at [time]. Review their changes before saving.” They can reload or copy their text. Enquiry status updates show who changed it last; the latest change wins and both are in the history.

G-08 What must be preserved as a historical record when a business object changes?

> For enquiries: every status change, note, who and when. For content: previous versions (at least the last 10), publish and archive times, publisher. For fee schedules and merit lists: every published version, kept with its dates, because people may rely on them.

G-09 What are the exact rules for duplicate submissions, reversals, cancellations, and resubmissions?

> - Same token resubmitted: ignored (Q-03).
> - Same phone and program within 7 days: kept and flagged Possible duplicate.
> - Visitor cancelling an enquiry: by asking admissions; status becomes Not interested and the personal data is deleted if requested.
> - Staff mistakes are reversed by restoring a previous version (L-09).
> - No payments, so no financial reversals.

H Search, Filtering & Discovery

Only answer where browsing or finding records is required.

H-01 Which pages or objects require search, filtering, sorting, or pagination?

Where	Needs
Programs list	Filter (field, level), no pagination
Notices	Filter (program, type, session), newest first, 10 per page
News and events	Newest first, 10 per page
Staff: Enquiries	Search, filter, sort, 25 per page
Site-wide search	Could (release 1.1)

H-02 Which fields can be searched, and should matching be exact, partial, or typo-tolerant?

> Staff enquiry search: name and phone number, partial match, ignoring spaces and dashes in phone numbers. Public site search (if built): page titles and text, partial match; typo tolerance not required.

H-03 What filters, sort orders, and default results ordering are required?

List	Filters	Default order
Programs	Field: Pharmacy, Physical therapy, Nursing, Allied health, Computing, Business; level	Alphabetical
Notices	Program, type, session	Newest first
Enquiries (staff)	Status, program, date range, city, source, possible duplicate	Newest first; can sort by name or status

H-04 What records must be hidden or excluded based on user permissions?

> Visitors never see drafts, archived-only items outside the archive, withdrawn programs or any enquiry. Content editors never see enquiries. Merit-list PDFs are public but excluded from search engines (O-13).

H-05 How should no-results and search-service failure states be shown?

> No results: “No matching notices. Clear filters or contact admissions.” with a Clear filters button. Failure: “Search isn’t working right now. Browse all notices” with a link to the unfiltered list.

H-06 Can users save searches, export results, or share filtered views?

> Visitors can share a filtered view because filters are in the page address. No saved searches. Staff can export the filtered enquiry list to Excel/CSV (with the confirmation in E-11).

H-07 How recent must searchable data be after an item is created or updated?

> Published content appears in public lists within 5 minutes. New enquiries appear in the staff list within 1 minute.

I Notifications & Communication

Specify triggers, channels, content requirements, and delivery rules.

I-01 What actions or events must trigger notifications?

- Enquiry or contact form received.
- Staff invited; password reset requested; password changed; two-factor reset; new-device sign-in (Could).
- Form-service or site failure (to the administrator).
- Daily summary of new enquiries (Could).

I-02 Who receives each notification, through which channel (email, SMS, in-app, etc.)?

Event	Recipient	Channel
Enquiry received	Admissions mailbox	Email
Enquiry received	Visitor (if email given)	Email auto-reply; Thank-you page on screen
Contact message	Admissions mailbox	Email
Account events	The staff member	Email
Failures, downtime	Administrator (developer)	Email; uptime alert by email or app
Daily summary (Could)	Admissions officer	Email

SMS: not used in release 1.

I-03 What information must each notification contain, and which sensitive details must it omit?

Admissions email: name, phone, email, program, city, qualification, marks, message, received time, source page, a link to the enquiry in the staff area. Visitor auto-reply: a thank-you, the program chosen, what happens next, office hours, WhatsApp and phone; it must **not** repeat the visitor's phone number or marks. Account emails never contain passwords.

I-04 What delivery deadlines or urgency levels apply to each notification?

Enquiry emails: within 1 minute. Account emails: within 2 minutes. Failure alerts: within 5 minutes of detection. Daily summary: 9:00 am PKT.

I-05 Can users opt in, opt out, mute, or configure notifications?

Visitors: no marketing messages exist, so nothing to opt into; a visitor can ask not to be contacted again (M-04). Staff cannot turn off security emails; admissions may turn the daily summary on or off.

I-06 What should happen when a notification cannot be delivered?

The enquiry is still stored, so nothing is lost. Bounced admissions email: the administrator is alerted and admissions checks the staff list instead. Bounced visitor auto-reply: ignored; admissions phones the visitor anyway.

I-07 Are reminders, digests, escalation messages, or scheduled alerts required?

Could: daily digest of new enquiries; reminder to admissions when an enquiry has stayed New for more than 1 working day. Not required for launch.

I-08 **Must notifications have read/unread status, history, or delivery evidence?**

> Not required. The enquiry record itself is the evidence: it shows when it arrived and when its status changed.

J External Services & Integrations

Define contracts at the behavioral level; the API specification comes later.

J-01 **Which external services are required for release 1 and what is each used for?**

Service	Use
Form-and-email service (OD-06)	Receive, store and email enquiries; send auto-replies
Email provider (OD-04)	Admissions mailbox and sender addresses
WhatsApp	Click-to-chat
Google Maps	Map and directions
Google Analytics 4 and Search Console	Measurement and search health
YouTube	Embedded videos
Spam protection (e.g. CAPTCHA)	Block bots on forms
Uptime monitor	Alert when the site is down

J-02 **Which user actions or scheduled events trigger each external service call?**

> Form submit → form service and spam check. Tap WhatsApp → WhatsApp. Open Contact or tap directions → Maps. Page view and button taps → Analytics (after the visitor allows analytics, if consent is required). Uptime monitor checks every 5 minutes.

J-03 **What information must be sent to or received from each service?**

> Form service: the enquiry fields from F-02; receives a success or failure result. WhatsApp: the institute number and a pre-filled message naming the program; nothing personal is sent by the site. Analytics: page address, button names and campaign tags only — never form values. Maps: the campus location only.

J-04 **What authentication, authorization, or consent is required for those transfers?**

> Service keys are kept in the hosting or tool settings, never in public page code where avoidable. The consent checkbox covers sending enquiry data to the form service. Analytics needs a privacy notice and, if required, a consent choice (O-08).

J-05 **What response and failure behavior is expected if a service is slow or unavailable?**

> Form service: wait up to 15 seconds, then show the failure message with WhatsApp and phone (Q-07). Maps: show address and a text link. YouTube: show a link. Analytics or spam check failing must never block the page or the form (spam check failure falls back to the hidden-field guard and rate limit).

J-06 **When should failed external actions be retried, queued, or escalated?**

> The visitor may retry a failed form submission manually. If the form service reports 3 failures within 1 hour, the administrator is alerted. No automatic queueing in release 1.

J-07 What must happen when an external service returns duplicate, incomplete, or invalid data?

> An enquiry that arrives incomplete is still kept and marked “Incomplete — check by phone”. Duplicates are flagged (F-05). An unexpected response from the form service is treated as a failure and shown to the visitor as such.

J-08 Who owns third-party accounts, credentials, contracts, and service limits?

> The **institute** owns every account (domain, hosting, email, form service, Google, WhatsApp number), registered to an institute email address, with at least two named people able to access it. The developer manages them day to day. Credentials are kept in a password manager owned by the institute. Free-plan limits (e.g. monthly form submissions) are checked before launch.

J-09 Is there a manual fallback if an essential external service is unavailable?

> Yes. Phone, WhatsApp and email addresses are shown beside every form, and walk-in enquiries continue on campus. If the form service is down for more than 1 hour, the form is replaced by a notice with these contacts.

K Payments, Transactions & Commercial Rules

Skip this section if the website handles no commercial transactions.

K-01 Will the website collect payments, issue refunds, or manage paid subscriptions?

> **Not applicable** for release 1. The website collects no payments. Fees are shown for information only.

K-02 Which payment methods, currencies, and customer regions must be supported?

> **Not applicable**. Fees are displayed in Pakistani rupees (PKR). A later phase may add JazzCash, Easypaisa or bank transfer.

K-03 What pricing, discounts, taxes, fees, and rounding rules must be applied?

> **Not applicable**. Fee tables show whole rupees, with thousands separators (e.g. Rs 125,000), as approved by admissions.

K-04 What are the states of a transaction and the exact rules for confirmation?

> **Not applicable**

K-05 What happens on failed, pending, cancelled, duplicated, or disputed payments?

> **Not applicable**

K-06 What receipts, invoices, and transaction histories must users receive?

> **Not applicable**

K-07 Who can approve a refund, and what conditions and deadlines apply?

> **Not applicable** on the website. Refunds stay under the institute’s existing policy.

K-08 Which billing information must be stored or must never be stored on the website?

> No card, bank-account or billing information may be collected or stored on the website. Official bank details for fee deposit may be **displayed** if admissions wants (OD-18), together with a warning against paying into personal accounts.

L Administration, Moderation & CMS

Describe operational controls, including staff-only behavior.

L-01 Which admin dashboards, views, and tools are required?

- Dashboard: new enquiries today and this week, enquiries by program, items waiting for review.
- Enquiries list and detail (search, filter, status, notes, export).
- Content lists: notices and merit lists, news and events, pages, programs, fees and dates.
- Media and files library.
- Users and roles; activity log (administrator).

The tool that provides these is **Not decided** (OD-05): a lightweight content-management tool or, at minimum, a shared spreadsheet for enquiries plus a simple editor for content.

L-02 What actions can each staff role perform, and which require additional approval?

Role permissions are in C-02. Extra approval is required for: fee and date changes, merit lists, program additions or withdrawals, affiliation and approval documents, and any official statement. The publisher records the approver's name.

L-03 What content can staff create, edit, publish, unpublish, or archive?

Notices, merit lists, news and events, program pages, admission pages, fee schedules, intake dates, faculty list, campus and gallery, job openings, about and leadership messages, legal pages (administrator only). Everything published can be unpublished or archived by its role owner.

L-04 Is a draft-review-publish process required, and who may approve publication?

Content	Process	Approver
Fees, dates, merit lists	Draft → In review → Published	Principal or Director
Program pages, affiliation, about	Draft → In review → Published	Director
Notices (general), news, events	Draft → Published (review optional)	Publisher's own judgement; Principal for official statements
Legal pages	Draft → In review → Published	Director

If the tool cannot record approvals, the approver's name and date are written in the item's internal note.

L-05 How should user reports, flags, complaints, or moderation queues be handled?

There is no user-generated content to moderate. Problem reports and complaints arrive by email to the administrator or admissions, are acknowledged within 2 working days and logged (date, issue, action, closed date).

L-06 What bulk actions, imports, exports, and mass edits are permitted?

> Export enquiries to Excel/CSV (admissions officer, administrator). Bulk status change of enquiries (e.g. mark selected as Spam). Bulk upload of up to 20 images. No bulk import of enquiries or content in release 1.

L-07 What filters and status indicators are needed in operational queues?

> Enquiries: status chips (colour plus text), Possible duplicate flag, age (“3 days old”), program, city, source. Content: Draft, In review, Published, Archived, and scheduled date.

L-08 What safeguard must appear before destructive or high-impact admin actions?

> Confirmation dialog naming the item for: delete, replace merit list, change fees or dates, export personal data, disable a user, change roles. Changing roles or disabling an administrator also asks for the administrator’s password.

L-09 What is the process for reversing an incorrect staff action?

> Restore the previous version of the content from its history (G-08), which republishes it and logs the restore. A wrong enquiry status is changed back and a note added. Deleted content can be recovered from backup within 14 days by the administrator.

L-10 Which administrative events must be auditable and visible to authorized reviewers?

> Sign-ins and failed sign-ins; user created, disabled, role changed; content published, edited, archived, deleted, restored; fee, date and merit-list changes; enquiry exports and deletions. Visible to the administrator; summaries shared with the Director on request.

M Data Requirements, Privacy Lifecycle & Migration

Specify required information and its lifecycle, not a detailed database schema.

M-01 What categories of data must the website collect, display, process, and maintain?

Category	Examples
Public content	Programs, fees, dates, notices, merit lists, news, faculty names and qualifications, photos
Enquiry data (personal)	Name, phone, email, city, qualification, marks, message, status, notes
Staff account data	Name, email, role, sign-in history
Operational data	Activity log, error logs, uptime data
Analytics	Anonymous page and event counts

M-02 Which data fields are classified as sensitive or personal?

> **Personal:** enquiry name, phone, email, city, message; staff names and emails. **Sensitive:** qualification marks; merit lists (contain students’ names, and sometimes marks or roll numbers); staff passwords and two-factor secrets. Release 1 collects no CNIC, B-form or date of birth.

M-03 What information must each role be able to view or export?

> Visitors: public content only. Admissions officer: all enquiry data, view and export. Content editor: no personal data. Administrator: everything, including logs; exports logged.

M-04 Which data should users be able to correct, download, or request removal of?

> A visitor may ask, by email or WhatsApp to admissions, to see, correct or delete their enquiry, or to stop being contacted. Admissions completes it within 14 days (proposed) and notes it in the enquiry history (keeping only "deleted on request, date"). A student may ask for a merit list correction through admissions; the corrected PDF replaces the old one.

M-05 How long must records, drafts, logs, and uploaded files be retained?

Data	Retention (proposed, OD-10)
Enquiries	12 months after the related intake closes
Contact messages	12 months
Drafts	Until published or deleted; flagged after 90 days
Notices and merit lists	Public for 2 years, then archived offline
Activity (audit) log	2 years
Error and server logs	90 days
Analytics	14 months
Backups	14 days

M-06 What events should trigger archival, anonymization, or permanent deletion?

- Intake closes + 12 months: enquiries deleted, or anonymised to program, city and status for statistics.
- Deletion request: personal fields deleted within 14 days.
- Notice archive date reached: moved to Archive.
- Notice older than 2 years: removed from the site and kept offline.
- Staff disabled for 90 days: account deleted; log entries keep the name.

M-07 What records require legal holds or exceptions to ordinary deletion rules?

> Records involved in a complaint, dispute or official inspection are kept until it is closed, on the Director's instruction. Whether any regulator requires longer retention of merit lists or admission records is **Not decided** (OD-15).

M-08 Where must data reside geographically, if there are residency requirements?

> No residency requirement is known for release 1; **Not decided** pending legal review (OD-15). Providers are chosen with reputable, secure data centres and the option to delete data on request.

M-09 Is existing data being migrated, and what sources/formats are involved?

> No automated migration: the old WordPress database is not accessible. Content is re-created by hand from approved sources: leadership messages and team list (old site copy), logo, photos and videos from the institute's social pages, and departments' program details. No enquiry or user data is migrated.

M-10 How should missing, corrupt, duplicate, or invalid imported data be handled?

> Each re-created item goes through the approval in L-04. Anything that cannot be confirmed (e.g. the online-MBA courses and statistics shown on the old test site) is left out and logged in the open decisions register. Template filler and the spam listing found on the old test site are never copied.

M-11 What reconciliation checks must prove that imported data is complete and correct?

> A content checklist (one row per page and program) signed by the relevant approver; a link check with no broken links; and a check that no placeholder text remains (search for "lorem", "demo", "example").

N Responsive UX, Accessibility & Localization

Record measurable interaction requirements across devices and audiences.

N-01 Which browsers, operating systems, screen sizes, and devices must be supported?

> Latest two versions of Chrome, Edge, Firefox, Safari (iOS) and Samsung Internet; Android 8+, iOS 15+, Windows 10 and 11, macOS. Screen widths 360 px to 1920 px; tested at 360, 390, 768, 1366 and 1920 px. Reference low-end device: an Android phone with 2–3 GB RAM. Internet Explorer: not supported.

N-02 What functions must work equally well on mobile, tablet, and desktop?

> All of them: reading every page, filters, downloads, the enquiry form, WhatsApp and call buttons (call shows the number on desktop), map, and the staff area (staff area must at least be usable on a phone for posting a notice).

N-03 What minimum accessibility standard is required (for example WCAG 2.2 AA)?

> WCAG 2.1 AA as the minimum (as agreed in the PVS), aiming for WCAG 2.2 AA where it adds no cost (e.g. target size, focus not obscured).

N-04 What keyboard navigation, focus handling, and visible focus indicators are required?

> - Every link, button, menu and form field reachable and usable by keyboard, in reading order.
> - A "Skip to content" link first on every page.
> - A clearly visible focus outline (at least 2 px, 3:1 contrast).
> - Menus and dialogs close with Esc; focus returns to the button that opened them.
> - After a form error, focus moves to the first field with an error.

N-05 What accessibility support is required for screen readers, labels, and error messages?

> - Correct page language, one main heading per page, headings in order, landmarks (header, nav, main, footer).
> - Every field has a visible label; required fields say “required” in text.
> - Errors are linked to their field and announced when they appear.
> - Images have meaningful alt text; decorative images are hidden from screen readers.
> - PDF notices also show their key details (title, date, program) as page text.

N-06 What contrast, text resizing, reduced-motion, and media caption requirements apply?

> Text contrast at least 4.5:1 (3:1 for large text and icons). Page usable at 200% zoom and at 320 px width without sideways scrolling. Animations are switched off when the device asks for reduced motion. Videos produced by the institute should have captions; embedded third-party videos at least a text summary.

N-07 Which languages and writing directions (LTR/RTL) must the website support?

> English, left-to-right, in release 1. Urdu, right-to-left with a proper Nastaliq or Naskh font, for key pages when translated (Could; OD-11). Layout must not break when Urdu text is entered in names or messages.

N-08 Which dates, times, numbers, addresses, and currencies must be localized?

> Dates as 17 Oct 2026; times as 2:30 pm, Pakistan time; phone numbers as 0300 1234567; currency as Rs 125,000 (PKR); addresses in Pakistani format. Urdu pages use the same numerals as the English pages unless the translator decides otherwise.

N-09 Can users select and retain language and appearance preferences?

> Language: once Urdu exists, an EN | اردو switch, remembered on the device. Appearance (dark mode): not required.

N-10 What must happen if a translation or localized resource is missing?

> Show the English page with a short note in Urdu: “یہ صفحہ ابھی صرف انگریزی میں دستیاب ہے” (this page is currently available only in English). Never show a blank page or raw text keys.

O Security, Identity Protection & Compliance

Specify security outcomes and controls that can be verified.

O-01 What user information or business operations have the highest security sensitivity?

> 1. Control of the domain and DNS (the site was already hijacked once).
> 2. Staff accounts, especially administrator.
> 3. Enquiry data (personal).
> 4. Publishing fees, dates and merit lists (false information harms students).
> 5. Hosting and email accounts.

O-02 Which actions require re-authentication, step-up verification, or dual approval?

Re-enter password: changing roles, disabling an administrator, changing own email or password, exporting enquiries. Two-factor at every sign-in for administrators. Dual approval: fee, date and merit-list publication (approver recorded, L-04). Domain and DNS changes: two named institute people informed before the change.

O-03 What rate limits or anti-abuse controls are required for login, signup, forms, and APIs?

Target	Control
Enquiry and contact forms	Spam check (CAPTCHA or equivalent) + hidden-field guard; at most 5 submissions per hour per device/IP
Staff sign-in	5 failed attempts → 15-minute lock; alert to administrator after 10 failures in an hour
Password reset	At most 3 requests per hour per account
Signup	**Not applicable** (no public signup)

O-04 What must happen after repeated failed login or verification attempts?

After 5 failures the account is locked for 15 minutes and the staff member is emailed. The message stays generic ("Email or password is incorrect") so it does not reveal whether the account exists. The administrator can unlock early.

O-05 Which data must be encrypted in transit and at rest?

In transit: everything, by HTTPS only (TLS 1.2 or newer), with HSTS. At rest: enquiry data, staff credentials (passwords stored only as strong one-way hashes), backups and exports, using the encryption the chosen providers offer. Exported files are kept only on institute devices and deleted after use.

O-06 What rules apply to cookies, session handling, CSRF, and cross-origin access?

- Visitors: no cookies except analytics, and only as allowed by the cookie choice (O-08).
- Staff session cookies: Secure, HttpOnly, SameSite=Lax or Strict; new session ID at sign-in; ended at sign-out and timeout (C-09).
- All staff forms protected against cross-site request forgery.
- Other websites cannot read data from the site (no open cross-origin access); the site cannot be framed by other sites.
- Security headers include a content security policy.

O-07 What file security controls are required for uploads and downloads?

Uploads only by staff; type checked by content; size limits (F-09); malware scan where available; stored so they can never run as code; cleaned filenames. Downloads served over HTTPS with the correct file type. Merit-list PDFs excluded from search engines.

O-08 What consent notices, privacy disclosures, and cookie preferences are required?

Privacy policy linked in the footer and under every form. Required consent checkbox on forms: "I agree that RIPS may contact me about admissions by phone, WhatsApp or email. See the privacy policy." A short cookie banner with Accept and Decline for analytics; analytics runs only after Accept (proposed, to be confirmed by legal review).

O-09 Which regulations, policies, or industry rules must the product comply with?

> Prevention of Electronic Crimes Act, 2016 (Pakistan); rules of the affiliating university, the Higher Education Commission and the relevant professional councils on advertising programs *(confirm)*; copyright law for images and text; WCAG 2.1 AA (A-08). Whether a Pakistani personal-data-protection law applies at launch: **Not decided**, legal counsel to confirm (OD-15).

O-10 What access-control checks must prevent users viewing another user's records?

> Visitors have no records to view. Every staff request is checked on the server for the user's role; hiding a menu item is not enough. Enquiry detail pages require the admissions or administrator role. Export files are never placed at public addresses.

O-11 Which security events require logging, alerting, or investigation?

Event	Response
10+ failed sign-ins in an hour	Alert administrator
Sign-in from a new device or country (Could)	Email the staff member
Role change, user created, export of enquiries	Log; weekly review
Site files changed outside a release	Alert administrator; investigate
DNS or nameserver change at DOMAIN.pk	Alert to two institute contacts
Google Search Console security warning	Alert administrator; investigate same day

O-12 What is the expected handling of suspected fraud, compromised accounts, or data breaches?

1. Disable the affected account or switch the site to the holding page.
2. Change passwords and keys; check DNS and nameservers.
3. Restore clean files from the last good backup.
4. Tell the Director at once; record what happened.
5. If personal data was exposed, inform affected people and any authority as the law requires.
6. For fake RIPS websites or fake admission offers: post a warning on the site and social pages and report the fake site to its host.

O-13 What is prohibited from appearing in responses, logs, analytics, and error messages?

> Passwords, reset links, two-factor codes, full enquiry contents in logs, phone numbers or names in page addresses or analytics, technical error details (stack traces, file paths, versions) on public pages, staff email addresses beyond the published contact address. Merit lists must not be indexed by search engines.

P Performance, Capacity & Reliability

State measurable expectations; do not prescribe implementation details unnecessarily.

P-01 How many daily, monthly, and concurrent users are expected at launch and after one year?

Not decided; no baseline exists (OD-13). Working assumption:

Measure	At launch	After one year
Visits per day (normal)	100–300	200–600
Visits per day (admission peak)	1,000–3,000	2,000–6,000
Visits per month	5,000–15,000	10,000–30,000
Concurrent visitors (peak)	100	200

P-02 What are the anticipated peak traffic scenarios and seasonal spikes?

Admissions opening; intermediate result day; merit-list publication (very sharp spike within minutes of a post on social media); final dates before closing. The site must handle 10× normal traffic for several hours.

P-03 What are the required loading-time targets for key pages and interactions?

On a mid-range Android phone over 4G: Largest Contentful Paint 2.5 s or less, page usable within 3 s, layout shift 0.1 or less, interaction response 200 ms or less. Page weight about 1.5 MB or less. Lighthouse mobile performance 85 or higher on Home, a program page, Admissions and Notices.

P-04 What response-time targets apply to search, submissions, and other critical tasks?

Filters update in under 1 s. Enquiry submission completes in under 5 s (95% of the time); over 15 s counts as failure (J-05). Staff saves under 3 s; published changes live within 5 minutes.

P-05 What are the maximum acceptable file upload and download times or sizes?

Public PDFs 5 MB maximum, ideally under 1 MB; each link shows its size. A 1 MB PDF should download in under 5 s on 4G. Staff upload of a 5 MB file under 30 s.

P-06 What availability or uptime target must the website meet?

99.5% per month (proposed), measured by an external uptime monitor checking every 5 minutes. During admission season, aim for 99.9%.

P-07 What maximum downtime and data loss are acceptable (RTO and RPO)?

RTO (time to restore): 4 hours; holding page within 30 minutes. RPO (data loss): 24 hours for site content (daily backup); **no loss** of enquiries, which are kept by the form service and emailed on arrival.

P-08 What service degradation is acceptable when traffic exceeds expectations?

Pages may load more slowly but must stay up; the merit-list PDF and notices are the priority. Non-essential items (video embeds, gallery) may be dropped. Enquiry form may show “busy, please try again or WhatsApp”.

P-09 **Which tasks must continue to work during a partial service outage?**

> Reading all public pages and downloading PDFs; WhatsApp and call buttons; contact details. Losing the form service or staff area must not take public pages down.

P-10 **What backup, restore, and business-continuity outcomes are required?**

> Daily automatic backup of site files and content, kept 14 days, stored separately from the host. Code kept in a Git repository with an off-site copy. Restore tested before launch and every 6 months. A ready-made holding page that can be switched on within 30 minutes.

P-11 **What response is expected if the user has a slow or unstable internet connection?**

> Text appears first; images load as the visitor scrolls; no autoplay video. If the form submission fails, typed values stay and the visitor can retry. PDF links show size so visitors know before downloading.

P-12 **What scalability milestones or load thresholds should prompt capacity changes?**

> Review hosting if: peak concurrent visitors exceed 300; monthly visits exceed 50,000; LCP goes over 3 s for a month; or the form service nears its monthly submission limit (alert at 80%).

Q Errors, Edge Cases & Recovery

Describe what the user and system should do when something goes wrong.

Q-01 **What should users see on expired sessions, blocked access, and invalid links?**

> Expired staff session: “Your session has ended. Please sign in again.” then return to the same page. Blocked: “You do not have permission to do this.” Invalid link: the 404 page (D-06). Expired reset link: “This link has expired. Request a new one.”

Q-02 **How should the website respond when required information is missing or malformed?**

> Field messages from F-07, the form is not submitted, focus moves to the first error. Content missing on a page (e.g. no fee entered): the page shows “Contact admissions for current fees” rather than a blank.

Q-03 **What must happen when a user double-clicks Submit or repeats an interrupted action?**

> The button is disabled after the first click. Each form carries a one-time token; a second submission with the same token is ignored and the visitor sees the same Thank-you page.

Q-04 **How should network loss during editing, payment, or submission be handled?**

> Visitor form: values kept, failure message shown, retry allowed. Staff editor: changes kept in the browser and the editor warns “Not saved — you appear to be offline.” Payment: **Not applicable**.

Q-05 **How should conflicts be handled when data changes after a page loads?**

> Staff: the conflict warning in G-07. Visitors: if a program closes after the page loaded, the enquiry is still accepted and admissions explains.

Q-06 **When can a failed action be retried safely, and when must it be blocked?**

> Safe to retry: form submission (protected by the token), downloads, staff saves. Blocked: sign-in during a lock, an expired reset link, and publishing a merit list that was already published for the same program and session unless “Replace” is chosen.

Q-07 How should the system handle third-party timeouts or inconsistent responses?

> Treat any timeout over 15 s or unexpected reply as a failure, show the fallback (J-05), never show a false success, and alert the administrator after 3 failures in an hour.

Q-08 What recovery path is offered when a user cannot complete a required step?

> Every error message includes WhatsApp, phone and email so a visitor can always reach admissions. Staff who are locked out contact the administrator, who confirms identity and resets access.

Q-09 Which failures must notify staff rather than just show an error to the user?

> Form-service failures (3 in an hour), site down, security warnings, DNS changes, failed backups, admissions mailbox bounces, and repeated failed sign-ins. All go to the administrator; form failures also go to admissions.

R Logging, Audit, Analytics & Monitoring

Define necessary evidence of what happened and product measurement.

R-01 Which user and staff actions must produce immutable or protected audit records?

> The staff events listed in L-10, plus enquiry status changes, deletions on request and exports. Staff cannot edit or delete audit records; only the administrator can read them.

R-02 What details should an audit event contain (actor, action, record, timestamp, outcome)?

> Who (staff name and role), what action, which item (title or enquiry reference), when (date and time, PKT), outcome (success or failure), and for changes the old and new value of key fields (status, fee, date). IP address only for sign-in events.

R-03 Who can view, search, and export audit history?

> Administrator: view, search, export. Director: receives exports on request. No one else.

R-04 What product analytics events must be tracked (such as signup, conversion, drop-off)?

Event	Purpose
page_view (with program name)	Interest per program
enquire_click	Intent
form_start, form_submit_success, form_error (field name only)	Conversion and drop-off
whatsapp_click, call_click, map_click	Contact intent
file_download (file name)	Prospectus and fee interest
notice_open	Notice use
outbound social click	Social engagement

R-05 Which operational metrics must be shown on dashboards?

> Uptime and response time; enquiries per day and by program; form failures; page speed (Core Web Vitals); Search Console clicks, impressions and errors; backup success.

R-06 **What alerts should be raised when services fail or critical thresholds are crossed?**

> Site down for 5 minutes; form-service failures 3 in an hour; security warning; DNS change; backup failed; SSL certificate expiring within 14 days; domain renewal due within 30 days; form-service quota at 80%.

R-07 **How long should analytics, diagnostic logs, and audit records be retained?**

> Analytics 14 months; diagnostic logs 90 days; audit records 2 years (M-05).

R-08 **What safeguards must stop sensitive personal information being sent to analytics tools?**

> - Form field values are never sent to analytics; only event names.
> - No personal data in page addresses (the Thank-you page address carries no name or number).
> - IP anonymisation and data-sharing settings off in Google Analytics.
> - Analytics only after consent where required (O-08).
> - Checked before launch by inspecting what the analytics tool receives.

S Interfaces, Environments & Operational Dependencies

Record externally visible environment requirements, ownership, and release dependencies.

S-01 **What environments are needed for development, testing, staging, and production?**

Environment	Where	Purpose
Development	Developer's computer (VS Code)	Building and first tests
Staging	Password-protected subdomain, proposed staging.rips.edu.pk	Review by Director, Principal, admissions; acceptance testing
Production	rips.edu.pk	Public site

S-02 **What behavior must be different in staging versus the public production website?**

> - Staging needs a password and tells search engines not to index it.
> - A visible "STAGING — not the real website" banner.
> - Forms send to a test mailbox, not admissions.
> - Analytics off.
> - May hold draft content not yet approved.

S-03 **Which domain names, redirects, and public URLs must exist at launch?**

> https://rips.edu.pk as the main address; http and www redirect to it. Public page addresses as listed in D-01. Known old WordPress addresses redirect permanently (D-05). sitemap.xml and robots.txt published. test.rips.pk removed or hidden (OD-17).

S-04 **What email sender addresses, SMS identities, and other external identities are required?**

> Proposed: **admissions@rips.edu.pk** (receives enquiries; replies), **no-reply@rips.edu.pk** (auto-replies and account emails), **webmaster@rips.edu.pk** (alerts, problem reports). Email authentication (SPF, DKIM, DMARC) set up so emails don't land in spam. Email provider **Not decided** (OD-04). One WhatsApp number for admissions (OD-14). No SMS identity in release 1.

S-05 Which service providers and assets must be available before the website can launch?

> Hosting (OD-03); DNS access at DOMAIN.pk; email provider and mailboxes (OD-04); form service (OD-06); Google Analytics and Search Console under an institute account; WhatsApp number; uptime monitor; backup storage; logo and photos; all approved content (A-05, M-11).

S-06 What manual operating procedures or staff training are needed for release 1?

> - One-page guides: posting a notice or merit list; updating fees and dates; handling an enquiry from New to Enrolled; handling a data-deletion request.
> - One-hour training session for admissions officer and content editor.
> - Administrator runbook: backups and restore, switching on the holding page, renewing domain and certificate, unlocking accounts, incident steps (O-12).

S-07 What is the required rollback behavior if a production release fails?

> Each release is tagged in Git. If a release breaks the site, redeploy the previous tagged version within 15 minutes; if that fails, switch on the holding page within 30 minutes. Content and enquiries are not lost by a rollback.

S-08 What support channels, escalation contacts, and response expectations are required?

Issue	Contact	Response
Visitor questions	Admissions (phone, WhatsApp, email)	Within 1 working day
Site down, hacked, wrong merit list published	Developer, then Director	Same day; start within 2 hours during office hours (proposed)
Normal bugs and content requests	Developer	Within 3 working days
Domain or hosting account issues	Administrator with DOMAIN.pk or host support	Same day

T Verification, Acceptance & Change Control

Define proof that the product meets its approved requirements.

T-01 What unique identifier format should each final functional and non-functional requirement use?

> Functional: **FR-[feature]-[nn]**, e.g. FR-003-04 (fourth requirement of the enquiry form). Non-functional: **NFR-[category]-[nn]** with categories PERF, SEC, AVL, ACC, PRIV, SEO, OPS, e.g. NFR-PERF-01. Features: FEAT-001…006. Open decisions: OD-01… Questionnaire IDs (A-01…T-10) are kept for traceability.

T-02 Which stakeholder accepts each major feature, and what evidence is needed?

Feature	Accepts	Evidence
FEAT-001 Program pages	Principal	Staging walkthrough; content checklist signed
FEAT-002 Admissions and fees	Admissions officer + Director	Figures match the approved fee sheet
FEAT-003 Enquiry form	Admissions officer	20 test submissions all received; auto-reply received
FEAT-004 Notices and merit lists	Admissions officer	Test notice and merit list published and archived
FEAT-005 Contact and WhatsApp	Admissions officer	Tested on Android and iPhone
FEAT-006 Staff management	Director	Staff publish unaided after training
Quality requirements (Worksheet 2)	Developer, reviewed by Director	Test reports

T-03 Which critical user journeys must pass end-to-end testing before launch?

1. Home → program → enquiry → Thank-you → admissions email → auto-reply.
2. Home → Admissions → download prospectus and fee PDF.
3. Home → Notices → filter → open merit list.
4. WhatsApp and call from a phone.
5. Staff sign-in with two-factor → publish notice → visible publicly → archive.
6. Admissions: enquiry New → Contacted → Enrolled → export.
7. Deletion request handled end to end.

T-04 What minimum browser/device/accessibility tests must pass?

Journeys 1–4 on Android Chrome (low-end and mid-range phone), iPhone Safari, Windows Chrome and Edge, and Samsung Internet. Automated accessibility check (e.g. axe) with no serious or critical issues; manual keyboard-only run of journeys 1 and 3; screen-reader spot check with TalkBack or VoiceOver; 200% zoom check.

T-05 What minimum security, performance, reliability, and recovery checks must pass?

- SSL Labs grade A; security-headers grade A or B; no high or critical findings in a web vulnerability scan.
- Malware scan clean; no Google security warning.
- Lighthouse mobile performance 85 or higher on the four key pages; LCP 2.5 s or less.
- Uptime monitor running; alert test received.
- Backup restored successfully to staging; rollback rehearsed.

T-06 How should failed tests, open defects, and acceptable known limitations be classified?

Class	Meaning	Launch rule
Critical	Security risk, data loss, a journey in T-03 broken, wrong official information	Blocks launch
High	Feature works with a hard workaround; serious accessibility problem	Fix before launch unless the approver accepts in writing
Medium	Minor feature fault with easy workaround	May launch; fix within 2 weeks
Low	Cosmetic	Backlog
Known limitation	Agreed out of scope (B-02)	Listed in release notes

T-07 Which issues must block launch regardless of schedule?

- Any security warning, malware or the old server still reachable at the domain.
- Enquiry form not delivering to admissions.
- Wrong fees, dates, eligibility or affiliation.
- Placeholder or template text anywhere.
- Broken mobile layout on the key pages.
- No privacy policy or consent.
- Institute email (MX) not working after the switch.
- Staff unable to sign in or publish a notice.
- No working backup.

T-08 What operational readiness, content, domain, support, and training checks must pass?

Content checklist signed (M-11); DNS, redirects and HTTPS verified; email sending and receiving tested; Search Console verified and sitemap submitted; staff trained and guides handed over (S-06); credentials stored in the institute password manager with two named owners; support contacts published (S-08); holding page ready.

T-09 What is the formal sign-off and release approval process?

1. Developer completes tests and the acceptance checklist on staging.
2. Feature owners sign their features (T-02).
3. Approver (OD-01) reviews the evidence pack (T-03 to T-08) and signs.
4. Go-live in a quiet hour; smoke test journeys 1–5 on production within 1 hour.
5. Approver confirms launch; release notes saved.

T-10 How will requirement changes after approval be evaluated, documented, and retested?

Each change is logged as CR-nn with requester, reason, affected requirement IDs, effort and risk. The approver accepts, defers or rejects it. Accepted changes update this SRS (new version number), and every affected test in T-03 to T-05 is rerun before release. Urgent security fixes may go out first and be documented within 2 working days.

REUSABLE WORKSHEETS

Worksheet 1 / Functional feature specifications

One specification per core feature. Acceptance tests use Given / When / Then.

FEAT-001: Program pages

Feature ID / name	**FEAT-001** — Program pages
Priority / release	Must · release 1
Description / user benefit	A list of programs and one page per program, so a student or parent can understand the program, check eligibility and fee, and enquire.
Actors / authorized roles	Visitor (read); content editor and administrator (edit); Principal approves content.
Trigger and preconditions	Visitor opens Programs or a program link. Program approved and affiliation confirmed; content entered.
Input data and validations	Visitor: filter choices only. Staff: name, level, duration, eligibility, fee, affiliation text, outline, careers, intake dates; all required except careers.
Normal step-by-step flow	1. Visitor opens Programs. 2. Site lists active programs alphabetically with field filter. 3. Visitor opens a program. 4. Site shows all fields, intake badge and last-updated date. 5. Visitor taps Enquire; form opens with the program pre-selected.
Alternative flows and cancellations	Visitor filters by field; prints the page; shares the link; taps WhatsApp instead of the form.
Status / data changes	None for visitors. Staff edits create a new version and an activity entry.
Errors and user-facing messages	Unknown program address: 404 page. Withdrawn program: “This program is not currently offered.” with link to Programs. Missing fee: “Contact admissions for current fees.”
Notifications / side effects	Analytics page_view with program name.
Postconditions / results	Visitor has the program facts and an Enquire path.
Acceptance tests (Given / When / Then)	1. **Given** an approved, active program, **when** a visitor opens its page on a 360 px phone, **then** name, level, duration, eligibility, fee, affiliation, intake badge and last-updated date are visible and the Enquire button shows without horizontal scrolling. 2. **Given** a program page, **when** the visitor taps Enquire, **then** the enquiry form opens with that program already selected. 3. **Given** a withdrawn program, **when** a visitor opens its old address, **then** the message “This program is not currently offered” appears with a link to Programs. 4. **Given** the Programs list, **when** a visitor chooses the Pharmacy filter, **then** only pharmacy programs are listed and the address includes the filter.

FEAT-002: Admissions and fees

Feature ID / name	**FEAT-002** — Admissions and fees
Priority / release	Must · release 1
Description / user benefit	Steps, required documents, dates, scholarships and fee tables for the current intake, plus prospectus and fee PDFs.
Actors / authorized roles	Visitor (read); admissions officer and administrator (edit); Principal or Director approves fees and dates.
Trigger and preconditions	Visitor opens Admissions. Current dates and fees approved and entered.
Input data and validations	Staff: intake opening and closing dates (closing after opening), fee lines per program in whole rupees, documents list, scholarship text, PDFs (≤5 MB), approver name.
Normal step-by-step flow	1. Visitor opens Admissions. 2. Site shows intake status, steps, documents, dates. 3. Visitor opens Fee structure; site shows fee table per program with last-updated date. 4. Visitor downloads prospectus or fee PDF.
Alternative flows and cancellations	Intake closed: page shows Closed and the next expected intake if known. Visitor goes straight to Enquire or WhatsApp.
Status / data changes	Intake status recalculated from dates daily. Fee changes create a new version kept in history.
Errors and user-facing messages	PDF missing: link hidden, “Contact admissions for the prospectus.” Staff enters closing date before opening date: “Closing date must be after the opening date.”
Notifications / side effects	file_download analytics event.
Postconditions / results	Visitor knows how, when and what it costs to apply.
Acceptance tests (Given / When / Then)	1. **Given** approved intake dates, **when** today is 7 days or fewer before the closing date, **then** the badge reads “Closing soon”. 2. **Given** an approved fee sheet, **when** the admissions officer compares it with the Fee structure page, **then** every amount matches and the last-updated date is shown. 3. **Given** a fee change, **when** the admissions officer saves it, **then** a confirmation asks for the approver’s name, and the previous version is kept in history. 4. **Given** the Admissions page, **when** a visitor taps the prospectus link, **then** the PDF downloads and its size was shown on the link.

FEAT-003: Enquiry form

Feature ID / name	**FEAT-003** — Enquiry form
Priority / release	Must · release 1
Description / user benefit	A short form that sends a visitor's details to admissions at any hour, so they are called back within 1 working day.
Actors / authorized roles	Visitor (submit); admissions officer and administrator (read, update, export).
Trigger and preconditions	Visitor taps Enquire or opens Contact. Form service and admissions mailbox working; privacy policy published.
Input data and validations	As F-02 and F-03: name, mobile, program, city and consent required; email, qualification, marks, message optional; spam guard; one-time token.
Normal step-by-step flow	1. Show form (program pre-selected if known). 2. Validate each field on exit. 3. On Submit, validate all fields and the spam guard. 4. Disable button, show "Sending...". 5. Send to form service; store with status New. 6. Email admissions; send auto-reply if email given. 7. Show Thank-you page.
Alternative flows and cancellations	Visitor closes the page: nothing stored. Chooses WhatsApp instead. Chooses "Not sure yet" for program. City "Other": types city.
Status / data changes	New enquiry record (status New, received time PKT, source page, campaign tag); flagged Possible duplicate if same phone and program within 7 days.
Errors and user-facing messages	Field messages from F-07. Service failure or timeout over 15 s: "We couldn't send your enquiry. Please try again, or WhatsApp us at [number]." Spam: silently discarded.
Notifications / side effects	Email to admissions within 1 minute; auto-reply to visitor; form_submit_success analytics event (no field values).
Postconditions / results	Admissions has the enquiry; visitor knows what happens next.
Acceptance tests (Given / When / Then)	1. **Given** a visitor on a program page, **when** they submit valid required fields with consent ticked, **then** the Thank-you page appears, admissions receives an email within 1 minute and the record has status New. 2. **Given** the form, **when** the mobile number is entered as 12345, **then** the message "Please enter a mobile number like 0300 1234567." appears and the form is not sent. 3. **Given** a filled form, **when** the visitor double-clicks Submit, **then** exactly one enquiry is stored. 4. **Given** the form service is unreachable, **when** the visitor submits, **then** the failure message with WhatsApp and phone appears and the typed values remain. 5. **Given** an enquiry from 0300 1234567 for Pharm-D 2 days ago, **when** the same number enquires about Pharm-D again, **then** the new enquiry is stored and flagged Possible duplicate.

FEAT-004: Notices and merit lists

Feature ID / name	**FEAT-004** — Notices and merit lists
Priority / release	Must · release 1
Description / user benefit	One official, dated list of notices and merit-list PDFs that students can filter by program and session.
Actors / authorized roles	Visitor (read); admissions officer (all notices and merit lists); content editor (general notices only); administrator; Principal approves merit lists.
Trigger and preconditions	Visitor opens Notices; staff clicks Add notice. Approved PDF ready; staff signed in with the right role.
Input data and validations	Title 5–120 chars; type; program (default All); session (required for merit lists); publish date (default today); archive date (after publish date); PDF ≤5 MB, real PDF; approver name for merit lists.
Normal step-by-step flow	1. Staff adds notice and uploads PDF. 2. System checks file and shows preview. 3. Staff publishes (merit list: confirmation naming program and session). 4. Notice appears at top of the public list within 5 minutes. 5. On archive date it moves to Archive.
Alternative flows and cancellations	Save as Draft; schedule a future publish time (Could); replace an existing merit list for the same program and session; cancel before publishing (nothing published).
Status / data changes	Draft → Published → Archived; publisher, time, approver recorded; replaced files kept as versions.
Errors and user-facing messages	Wrong file type: “Please upload a PDF.” Too large: “The file is larger than 5 MB.” Content editor tries merit list: “Only admissions can publish merit lists.” Removed notice: “This notice is no longer available.”
Notifications / side effects	Activity log entry; merit-list PDFs marked not-for-search-engines; notice_open analytics event.
Postconditions / results	Official notice available at a stable, shareable address.
Acceptance tests (Given / When / Then)	1. **Given** a signed-in admissions officer, **when** they publish a merit list for Pharm-D 2026–27, **then** a confirmation names the program and session, and after confirming, the list appears first on Notices within 5 minutes. 2. **Given** a content editor, **when** they try to choose type Merit list, **then** the option is not available or the message “Only admissions can publish merit lists.” appears. 3. **Given** a published merit-list PDF, **when** a search engine requests it, **then** it is marked not to be indexed. 4. **Given** a notice whose archive date has passed, **when** a visitor opens Notices, **then** it no longer appears in the current list but is in Archive.

FEAT-005: WhatsApp, call and contact

Feature ID / name	**FEAT-005** — WhatsApp, call and contact
Priority / release	Must · release 1
Description / user benefit	One-tap WhatsApp chat and phone call, plus a contact page with address, hours and map.
Actors / authorized roles	Visitor; administrator maintains numbers and hours.
Trigger and preconditions	Visitor taps the WhatsApp button, call button or Contact. Staffed WhatsApp number, phone, verified address and office hours configured (OD-14).
Input data and validations	None from visitor. Pre-filled WhatsApp message: “Hello RIPS, I would like information about [program].” (program omitted if unknown).
Normal step-by-step flow	1. Visitor taps WhatsApp. 2. WhatsApp opens with the institute number and pre-filled message. 3. Visitor sends; admissions replies in office hours.
Alternative flows and cancellations	On desktop, call button shows the number; WhatsApp opens WhatsApp Web. Visitor opens map for directions.
Status / data changes	None on the website.
Errors and user-facing messages	WhatsApp not installed: WhatsApp Web or the number shown. Map unavailable: address text and an “Open in Google Maps” link.
Notifications / side effects	whatsapp_click, call_click, map_click analytics events.
Postconditions / results	Visitor is in direct contact with admissions.
Acceptance tests (Given / When / Then)	1. **Given** an Android phone with WhatsApp, **when** the visitor taps WhatsApp on a Pharm-D page, **then** WhatsApp opens to the admissions number with the Pharm-D message pre-filled. 2. **Given** an iPhone, **when** the visitor taps Call, **then** the dialer opens with the admissions number. 3. **Given** the Contact page, **when** the visitor taps Directions, **then** Google Maps opens at the verified campus location.

FEAT-006: Staff content and enquiry management

Feature ID / name	**FEAT-006** — Staff content and enquiry management
Priority / release	Should · release 1 (Must for notices and enquiries; if the editor tool slips, the developer publishes content on request and the full editor moves to release 1.1)
Description / user benefit	Staff publish content and track enquiries without the developer.
Actors / authorized roles	Admissions officer, content editor, administrator (permissions in C-02).
Trigger and preconditions	Staff signs in. Account active with two-factor; admin tool chosen (OD-05); staff trained.
Input data and validations	Sign-in: email, password, two-factor code. Content fields per F-02. Enquiry status from the allowed list (G-02) and optional note (≤500 chars).
Normal step-by-step flow	1. Staff signs in and lands on their role page (D-03). 2. Opens an item, edits, previews. 3. Saves as Draft or publishes (with confirmation where L-08 applies). 4. For enquiries: opens New enquiry, calls the person, sets status Contacted and adds a note. 5. Signs out.
Alternative flows and cancellations	Leave without saving: warning. Restore a previous version. Export filtered enquiries (with confirmation).
Status / data changes	Content versions and statuses; enquiry status history; activity log entries.
Errors and user-facing messages	Wrong credentials: “Email or password is incorrect.” Locked: “Too many attempts. Try again in 15 minutes.” No permission: “You do not have permission to do this.” Save failure: “Not saved — please try again.”
Notifications / side effects	Security emails; administrator alert on 10+ failed sign-ins per hour.
Postconditions / results	Public site reflects the change within 5 minutes; enquiry progress is recorded.
Acceptance tests (Given / When / Then)	1. **Given** a trained admissions officer, **when** they post a notice using the one-page guide without help, **then** it is live within 5 minutes. 2. **Given** a staff account, **when** 5 wrong passwords are entered, **then** the account is locked for 15 minutes and the staff member is emailed. 3. **Given** a content editor, **when** they open the Enquiries address, **then** access is denied. 4. **Given** a wrongly edited program page, **when** the editor restores the previous version, **then** the old content is live again and the restore is logged. 5. **Given** 10 enquiries filtered by program, **when** the admissions officer exports, **then** a confirmation warns about personal data and the file contains exactly those 10.

Worksheet 2 / Quality requirements

Measurable non-functional requirements. Targets marked in the answers as proposed still need the approver's confirmation.

ID and title	Category	Applies to	Measurement / test environment	Target	Owner · priority · evidence
NFR-PERF-01 Page speed	Performance	Home, program page, Admissions, Notices	Lighthouse mobile and PageSpeed Insights on production; mid-range Android on 4G	LCP ≤ 2.5 s; performance score ≥ 85; CLS ≤ 0.1	Developer · Must · reports
NFR-PERF-02 Page weight	Performance	All public pages	Browser network panel	≤ 1.5 MB first load	Developer · Must · screenshot
NFR-PERF-03 Peak load	Performance	Notices and merit-list PDF	Load test on staging	300 concurrent visitors with LCP ≤ 4 s and no errors	Developer · Should · test report
NFR-AVL-01 Availability	Reliability	Whole public site	External uptime monitor, 5-minute checks	≥ 99.5% per month	Administrator · Must · monthly report
NFR-AVL-02 Recovery	Reliability	Site content and files	Restore drill to staging	RTO 4 h (holding page 30 min); RPO 24 h; enquiries no loss	Administrator · Must · drill record
NFR-AVL-03 Form delivery	Reliability	Enquiry and contact forms	20 test submissions; monthly spot checks	100% in test; ≥ 98% in live use; email within 1 min	Admissions officer · Must · test log
NFR-SEC-01 Transport security	Security	All addresses	SSL Labs	Grade A; HTTPS only; HSTS	Developer · Must · report
NFR-SEC-02 Security headers	Security	All pages	securityheaders.com / Mozilla Observatory	Grade A or B	Developer · Must · report
NFR-SEC-03 Vulnerabilities	Security	Public site and staff area	Web vulnerability scan on staging	No high or critical findings	Developer · Must · scan report
NFR-SEC-04 Staff access	Security	Staff area	Manual test	2FA for administrators; lock after 5 failures; 30-min idle timeout	Developer · Must · test log
NFR-ACC-01 Accessibility	Accessibility	All public pages and forms	axe automated check + keyboard and screen-reader spot checks	WCAG 2.1 AA; no serious or critical issues	Developer · Must · report
NFR-PRIV-01 No personal data in analytics	Privacy	Analytics, logs, addresses	Inspect analytics debug view and logs	Zero names, phones, emails or marks found	Developer · Must · screenshots
NFR-SEO-01 Search visibility	SEO	Must-have pages	Search Console	All must-have pages indexed within 30 days; Lighthouse SEO ≥ 90; merit lists not indexed	Developer · Should · Search Console report
NFR-COMP-01 Browser support	Compatibility	Journeys T-03 1–5	Manual tests on devices in N-01	All pass	Developer · Must · test matrix

Open decisions register

Every "Not decided" in this document. Reviewed at each weekly meeting; status becomes Decided or Dropped with the date.

ID	Question / decision	Owner / due	Status
OD-01	Who is the single approver for requirements, scope and launch?	Chairman / before SRS sign-off	Open
OD-02	Launch date (before next intake)	Director / before development	Open
OD-03	New hosting provider	Director with developer / before development	Open
OD-04	Email provider and how MX records move	Director with developer / before DNS switch	Open
OD-05	Admin and content-editing tool	Developer, approved by Director / before development	Open
OD-06	Form-and-email service (and its plan limits)	Developer / before development	Open
OD-07	Final program list, exact degree titles, fees, eligibility	Principal and admissions / before content entry	Open
OD-08	Affiliation and approval documents and wording	Director / before content entry	Open
OD-09	Official spelling of the institute's name; use of "University"	Chairman / before design	Open
OD-10	Retention periods (enquiries, notices, logs)	Director / before launch	Open
OD-11	Urdu pages in release 1 or later	Director / before content entry	Open
OD-12	Budget for hosting, services and photography	Director / before development	Open
OD-13	Baseline figures (enquiries per intake, calls)	Admissions officer / before launch	Open
OD-14	Admissions WhatsApp number, phone and office hours	Admissions officer / before development	Open
OD-15	Legal review: data protection, retention, regulator rules	Director (legal counsel) / before launch	Open
OD-16	Staging address (proposed staging.rips.edu.pk)	Developer / before M2	Open
OD-17	Ownership of test.rips.pk; take down or hide it	Director with developer / immediately	Open
OD-18	Whether to show official bank details for fee deposit	Director and admissions / before launch	Open

SRS completion checklist

[ ] Every requirement has a clear owner and stable identifier.

[ ] Business rules, access permissions, and exceptions are testable.

[ ] Non-functional targets are measurable, not subjective.

[ ] Every essential feature has acceptance criteria and a named approver.

[ ] All unanswered questions are marked Not applicable or logged as open decisions.

> **Left unticked on purpose.** Identifiers, tests and the decision register are in place, but the approver (OD-01) is not yet named, so the fourth item cannot be met. The approver ticks these at sign-off.